

DATA ANALYTICS · PERSONAL PROJECT

Shopee App Review Analysis

Transforming 50,000+ Google Play Store Reviews into Strategic Business Insights

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85,500

Raw Reviews

50,611

Processed

8

Categories

2018–2024

Period

DISCLAIMER

Important Notice

This presentation is a personal project created solely for educational and portfolio purposes. The data used is a publicly accessible dataset obtained from Kaggle (<https://www.kaggle.com/datasets/ahmadseloabadi/shoppe-app-reviews-from-google-play-store>) and reflects only a specific time period covered by that dataset.

This project is not affiliated with, sponsored by, or representative of the internal data or official views of Shopee or any related companies. All findings, insights, and recommendations presented herein are based solely on the analyzed public dataset and are intended to demonstrate analytical capabilities and methodology.

Google Data Analytics Framework was applied throughout this project.

Background & Business Problem

BACKGROUND

- E-commerce success is driven by continuous digital evolution.
- App store ratings directly shape brand credibility and new user acquisition.
- Unstructured review data, when analyzed, becomes a strategic asset.
- Shopee receives thousands of reviews — manual analysis is impractical.

BUSINESS PROBLEM

Shopee receives thousands of app reviews on the Google Play Store, making manual analysis highly inefficient. The complexity increases when a single review addresses multiple service areas simultaneously.

Solution Needed:

- An intuitive, easy-to-understand dashboard
- Category-based complaint classification
- Time-series trend analysis
- Rating distribution monitoring

Project Objectives

Three clear goals to extract value from unstructured review data

01

Identify Priority Areas for Improvement

Determine which service categories — Shipping, App Performance, Customer Service, etc. — receive the highest volume of negative feedback and require the most immediate attention.

02

Analyze Complaint Trends Over Time

Track and quantify complaint volumes across months and years to reveal seasonal patterns, identify spikes, and measure the impact of platform updates on user satisfaction.

03

Determine Rating Distribution

Calculate the percentage breakdown of 1-to-5-star ratings to gauge overall user satisfaction levels, understand general sentiment, and benchmark platform health.

Dataset & Tools

DATASET

Shopee App Reviews — Google Play Store (Indonesia)

Source: Kaggle · Ahmad Seloabadi

85,500

Raw Records

11

Columns

Key Columns:

- reviewId, userName, userImage
- content (review text)
- score (1–5 stars)
- thumbsUpCount
- at (timestamp)
- reviewCreatedVersion, appVersion

TOOLS USED

Python & Pandas

Data manipulation, text processing & cleaning

Google Sheets / Excel

Data storage & bridge to visualization layer

Google Looker Studio

Interactive dashboard for stakeholder reporting

Methodology

Google Data Analytics Framework · 6 Phases

01 ASK

Define the business problem, stakeholder objectives, and key analytical questions.

02 PREPARE

Source the dataset from Kaggle and assess data integrity and relevance.

03 PROCESS

Clean and transform raw data using Python & Pandas — handle nulls, deduplicate, standardize.

04 ANALYZE

Categorize reviews, calculate rating distributions, and identify temporal trends.

05 SHARE

Build an interactive Google Looker Studio dashboard to communicate insights to stakeholders.

06 ACT

Formulate actionable, data-driven recommendations to guide service improvements.

Data Processing

Three-stage execution: *Cleaning* → *Transformation* → *Exploratory Analysis*



Data Cleaning



- Removed inconsistencies & empty rows
- Deduplicated records to prevent skewed metrics
- Reformatted timestamps into Month/Year format



Data Transformation



- Sentiment classification: 1–3★ = Negative, 4–5★ = Positive
- Keyword-based categorization into 8 service categories
- Metric aggregation: avg. rating, complaint volumes, sentiment %



Exploratory Analysis



- Year-over-year complaint trend analysis
- Rating distribution (1–5 star breakdown)
- Category prioritization by complaint volume

Key Findings

01 Sentiment & Rating Distribution

52.9%

Positive Sentiment

4–5 star reviews

47.1%

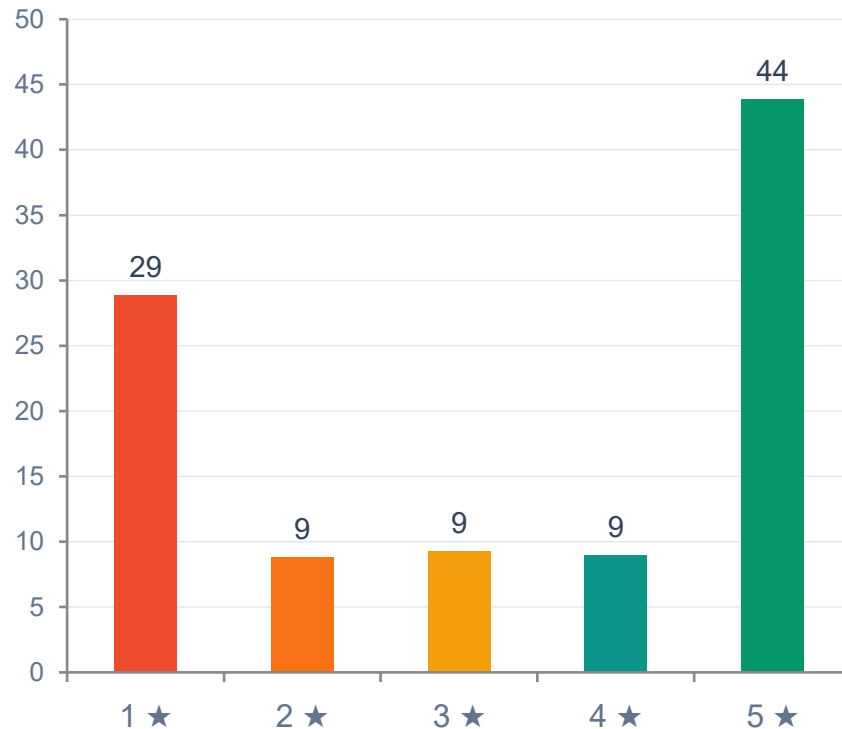
Negative Sentiment

1–3 star reviews

💡 Key Insight

User sentiment is notably polarized. While 5-star reviews dominate at 43.9%, a critical 28.9% of all reviews are 1-star — indicating a sizeable segment of highly dissatisfied users. Addressing this gap is essential for improving overall platform ratings.

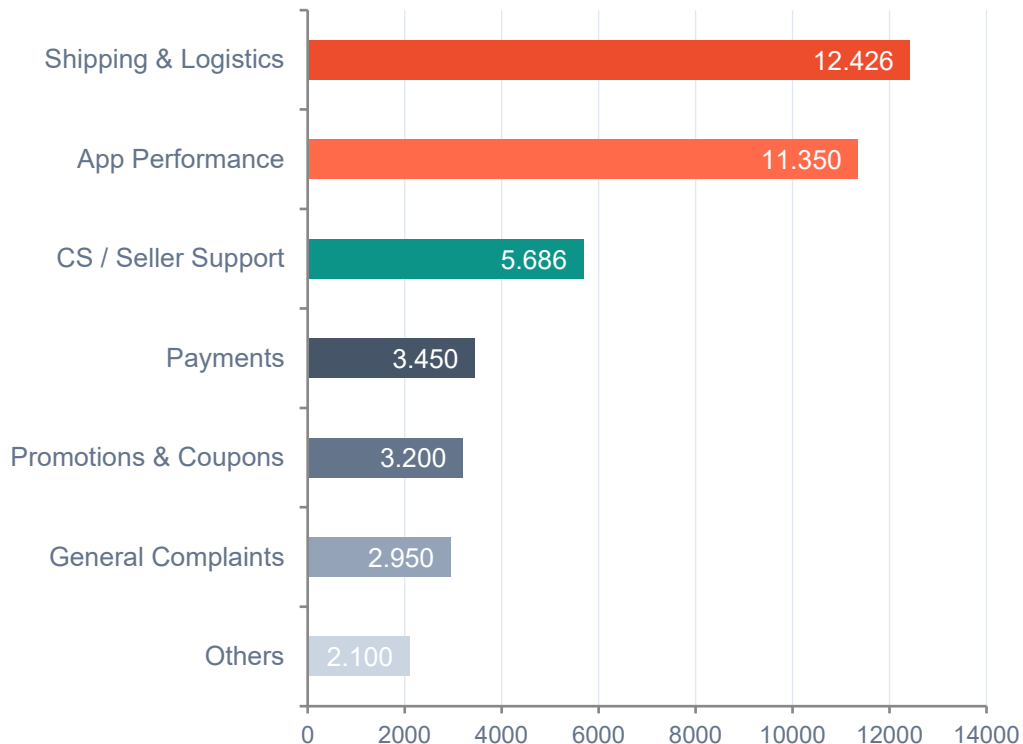
Rating Distribution (%)



Key Findings

02 Complaint Categories · 2018–2024

Total Complaints by Category (2018–2024)



TOP 3 PAIN POINTS

#1

Shipping & Logistics

12,426 complaints · 52.2% of all complaints

#2

App Performance

11,350 complaints · 47.6% of all complaints

#3

CS / Seller Support

5,686 complaints · 23.9% of all complaints

Total: 23,838 complaints out of 50,611 reviews (47.1%)

Key Findings

03 2024 Spotlight — Highest Feedback Year on Record

35,259

Total Reviews in 2024

Highest single-year volume

13,502

Complaints in 2024

38.3% of all-time complaints

March

Peak Complaint Month

517 complaints recorded

CATEGORY SHIFT IN 2024

1st

Shipping & Logistics



6,522

2nd

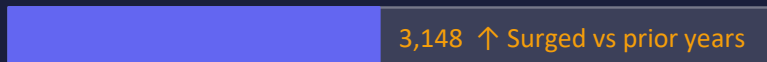
App Performance



4,726

3rd

General Complaints



3,148 ↑ Surged vs prior years

⚠ The rise of 'General Complaints' in 2024 may signal broader user fatigue or dissatisfaction with recent platform changes — requiring deeper UX investigation.

Recommendations

Three actionable priorities for Shopee's management and technical teams

R1

Optimize Shipping & Courier Operations

- Evaluate Shopee Express & third-party logistics partners
- Implement stricter SLA monitoring for couriers
- Improve real-time tracking transparency for users
- Provide faster resolution for lost or delayed packages

R2

Enhance App Stability & Performance

- Prioritize bug fixing and critical crash resolution
- Reduce app bloatware and optimize load times
- Conduct load testing before peak event months
- Focus on March — the peak complaint month

R3

Address the Surge in General Complaints

- Conduct deeper UX research into hidden friction points
- Investigate user fatigue from recent platform changes
- Launch re-engagement and loyalty campaigns
- Prevent user churn to competing platforms

Conclusion

This project demonstrates the value of data analytics in transforming unstructured consumer feedback into strategic business insights.

- Over 50,000 processed reviews were analyzed across 2018–2024.
- Shopee maintains a strong base of satisfied users (43.9% giving 5 stars).
- Critical bottlenecks in Shipping Logistics and App Performance are driving 28.9% of reviews to 1-star ratings.
- The developed Looker Studio dashboard enables stakeholders to dynamically monitor sentiment and pinpoint friction areas.
- By prioritizing the three key recommendations, Shopee can reduce negative sentiment, elevate its app rating, and drive sustainable user growth.

PROJECT IMPACT

50,611

Reviews Analyzed

23,838

Complaints Classified

8

Service Categories

3

Actionable Recommendations

Let's Connect

I'm open to data analyst opportunities, collaborations, and feedback.



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Thank You

for your time and consideration

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